

13 (a) Show that the equation $x^4 - 3x - 1 = 0$ has a solution between $x = 1$ and $x = 2$

(2)

(b) Show that for $x > 0$ the equation $x^4 - 3x - 1 = 0$ can be arranged to give $x = \sqrt[4]{3x + 1}$

(1)

(c) Starting with $x_0 = 1$, use the iteration formula $x_{n+1} = \sqrt[4]{3x_n + 1}$ once to find an estimate for a solution of $x^4 - 3x - 1 = 0$

.....
(1)

(Total for Question 13 is 4 marks)
